

切削参数参考表
Recommended Milling Conditions

加工材料 Work Material		铜合金 Copper						
外径 Dia.	颈长(t1) Under Neck Length	侧面 Side Milling				沟槽 Slotting		
		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm
0.1	0.5	40,000	140	0.1	0.004	40,000	130	0.007
	1	40,000	140	0.1	0.003	40,000	130	0.006
	1.5	40,000	140	0.1	0.002	40,000	130	0.005
	2	40,000	140	0.1	0.001	40,000	130	0.004
0.15	0.5	40,000	180	0.1	0.007	40,000	170	0.012
	0.6	40,000	180	0.1	0.006	40,000	170	0.01
	1	40,000	140	0.1	0.005	40,000	130	0.08
	1.5	40,000	140	0.1	0.004	40,000	130	0.07
	2	40,000	140	0.1	0.003	40,000	130	0.06
	2.5	40,000	140	0.1	0.002	40,000	130	0.05
0.2	1	40,000	350	0.2	0.006	40,000	320	0.015
	1.5	40,000	300	0.2	0.004	40,000	250	0.01
	2	40,000	300	0.2	0.003	40,000	250	0.08
	3	40,000	300	0.2	0.002	40,000	250	0.06
	4	40,000	300	0.2	0.001	40,000	250	0.05
0.25	1	40,000	400	0.2	0.01	40,000	380	0.03
	1.5	40,000	350	0.2	0.008	40,000	320	0.02
	2	40,000	300	0.2	0.006	40,000	250	0.015
	3	40,000	300	0.2	0.004	40,000	250	0.01
	4	40,000	300	0.2	0.003	40,000	250	0.008
0.3	1	40,000	500	0.3	0.01	40,000	450	0.035
	1.5	40,000	450	0.3	0.008	40,000	400	0.025
	2	40,000	380	0.3	0.006	40,000	350	0.017
	2.5	40,000	350	0.3	0.005	40,000	300	0.015
	3	40,000	350	0.3	0.004	40,000	300	0.013
	4	40,000	300	0.3	0.003	40,000	300	0.01
	6	40,000	300	0.3	0.002	40,000	300	0.008
	2	40,000	600	0.4	0.015	40,000	550	0.03
0.4	3	35,000	500	0.4	0.01	35,000	450	0.02
	4	28,000	350	0.4	0.006	28,000	300	0.015
	6	28,000	350	0.4	0.005	28,000	300	0.01
	2	38,000	800	0.5	0.02	35,000	700	0.055
0.5	3	35,000	700	0.5	0.015	32,000	600	0.04
	4	28,000	550	0.5	0.008	26,000	500	0.03
	5	18,000	500	0.5	0.007	18,000	300	0.02
	6	18,000	350	0.5	0.005	18,000	300	0.015
	8	18,000	350	0.5	0.004	18,000	300	0.01
	10	18,000	350	0.5	0.002	18,000	300	0.008
	12	18,000	350	0.5	0.001	18,000	300	0.006

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		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm
0.6	2	38,000	1,000	0.6	0.025	35,000	850	0.1
	3	32,000	800	0.6	0.02	30,000	700	0.08
	4	28,000	700	0.6	0.015	26,000	600	0.06
	6	20,000	450	0.6	0.01	20,000	400	0.03
	8	20,000	450	0.6	0.008	20,000	400	0.02
0.7	2	30,000	1300	0.7	0.05	28,000	1200	0.2
	4	26,000	1100	0.7	0.04	24,000	1000	0.15
	6	22,000	900	0.7	0.03	18,000	650	0.012
	8	16,000	600	0.7	0.02	16,000	500	0.01
0.8	2	30,000	1,500	0.8	0.05	28,000	1,400	0.15
	3	30,000	1,300	0.8	0.04	28,000	1,200	0.15
	4	26,000	1,100	0.8	0.03	24,000	1,000	0.12
	6	22,000	900	0.8	0.02	18,000	650	0.08
	8	16,000	600	0.8	0.01	16,000	500	0.05
1	10	16,000	600	0.8	0.008	16,000	500	0.05
	4	24,000	2,000	1	0.05	22,000	1,800	0.2
	6	20,000	1,500	1	0.03	18,000	1,200	0.14
	8	16,000	1,200	1	0.025	15,000	1,000	0.1
	10	14,000	1,000	1	0.02	12,000	800	0.07
1.5	12	10,000	700	1	0.01	10,000	650	0.05
	16	10,000	700	1	0.008	10,000	650	0.03
	6	18,000	2,200	1.5	0.08	16,000	1,800	0.3
	8	16,000	1,700	1.5	0.06	14,000	1,400	0.25
	10	16,000	1,500	1.5	0.05	14,000	1,400	0.2
	12	12,000	1,200	1.5	0.04	11,000	1,000	0.15
2	16	10,000	900	1.5	0.02	10,000	800	0.08
	20	10,000	900	1.5	0.01	10,000	800	0.06
	8	16,000	2,200	2	0.09	14,000	1,900	0.4
	10	14,000	1,900	2	0.08	12,000	1,600	0.35
	12	12,000	1,600	2	0.07	11,000	1,400	0.28
	16	10,000	1,200	2	0.045	9,000	1,000	0.18
备注 Notes	20	9,000	1,000	2	0.03	8,000	850	0.12
	※ 本切削参数仅供参考。请根据实际的加工形状和所使用的机床等调整切削参数。 ※ 切深量的ap表示轴向切入量，ae表示半径方向的切入量。 ※ 发生振刀时，请以相同的比率降低主轴转速和进给速度。 此外，主轴转速过低时，也以相同的比率调整。 ※ 建议使用油冷冷却方式进行冷却。 ※ Recommend to use the milling condition as just reference. Adjust milling conditions according to machining shape and machine status. ※ Depth of Cut : ap=Axial Depth of Cut / ae=Radial Depth of Cut. ※ Reduce both spindle speed and feed at same rate for chattering and also for insufficient spindle speed of a machine. ※ Water-insoluble cutting fluid is recommended.							

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